

B.13

$$X = x_2 x_1 x_0, \quad Y = y_2 y_1 y_0$$

$$\mathbf{X} = \mathbf{Y}$$

Just XNOR each pair of bits and AND them all together:

$$\text{Equal} = \overline{(x_2 \oplus y_2)} \cdot \overline{(x_1 \oplus y_1)} \cdot \overline{(x_0 \oplus y_0)}$$

$\mathbf{X} < \mathbf{Y}$ (Unsigned)

Check from the MSB and at the first bit where they differ, if $x_i = 0$ and $y_i = 1$, then $X < Y$.

Let $\text{LT}(x_i, y_i) = \bar{x}_i \cdot y_i$ and $\text{EQ}(x_i, y_i) = \overline{x_i \oplus y_i}$.

$$\text{UnsignedLT} = \text{LT}(x_2, y_2) + \text{EQ}(x_2, y_2) \cdot \text{LT}(x_1, y_1) + \text{EQ}(x_2, y_2) \cdot \text{EQ}(x_1, y_1) \cdot \text{LT}(x_0, y_0)$$

$\mathbf{X} < \mathbf{Y}$ (Signed)

Same idea, but the MSB comparison flips. If $x_2 = 1$ and $y_2 = 0$, X is negative and Y is positive, so $X < Y$. That's the opposite of the unsigned case. The lower bits stay the same.

$$\text{SignedLT} = x_2 \cdot \bar{y}_2 + \text{EQ}(x_2, y_2) \cdot \text{LT}(x_1, y_1) + \text{EQ}(x_2, y_2) \cdot \text{EQ}(x_1, y_1) \cdot \text{LT}(x_0, y_0)$$

Extension to 6 Bits

Split into a high group (bits 5,4,3) and low group (bits 2,1,0). Run the 3-bit comparator on each half.

$$\text{LessThan}_6 = \text{LT}_{\text{hi}} + \text{EQ}_{\text{hi}} \cdot \text{LT}_{\text{lo}}$$

$$\text{Equal}_6 = \text{EQ}_{\text{hi}} \cdot \text{EQ}_{\text{lo}}$$

Same pattern scales to any width. For 12 bits use two 6-bit comparators, etc.

B.14

Two inputs A, B . Two outputs C, D . Control S .

$S = 1$ (pass-through): $C = A, D = B$. $S = 0$ (crossing): $C = B, D = A$.

Each output is just a 2-to-1 mux:

$$\begin{aligned}C &= S \cdot A + \bar{S} \cdot B \\D &= S \cdot B + \bar{S} \cdot A\end{aligned}$$

Gate count: 1 NOT, 4 ANDs, 2 ORs.

Truth table:

S	A	B	C	D
0	0	0	0	0
0	0	1	1	0
0	1	0	0	1
0	1	1	1	1
1	0	0	0	0
1	0	1	0	1
1	1	0	1	0
1	1	1	1	1

$S = 0$: C follows B , D follows A . $S = 1$: C follows A , D follows B . Checks out.

B.24

The ALU uses the sign bit of $A - B$ for `slt`. Problem: this breaks when overflow happens.

The Problem

-7 (1001) minus 6 (0110):

$$1001 - 0110 = 1001 + 1010 = 0011$$

Sign bit is 0, so the ALU thinks $-7 \geq 6$. Wrong. The real answer is -13 , which overflows 4 bits.

Detecting Overflow

Overflow happens when the carry into the MSB differs from the carry out of the MSB:

$$\text{Overflow} = C_{\text{in,MSB}} \oplus C_{\text{out,MSB}}$$

The Fix

Use overflow to correct the sign bit:

$$\text{SLT} = \text{Sign} \oplus \text{Overflow}$$

No overflow: sign bit is already correct, XOR with 0 does nothing. Overflow: sign bit is flipped from the true answer, XOR with 1 flips it back.